

CSI5340 / ELG5214 Final Project Report

Deep Learning / Reinforcement Learning – Group Project

This document describes the expected structure, content, and grading breakdown for the Final Project Report. The final report must follow from your Project Proposal and reflect the research questions, methodology, and motivation you previously outlined. Deviations are allowed *only if clearly justified*.

The goal of the final report is to demonstrate your ability to:

- Execute a sound experimental research plan
 - Critically analyze results
 - Communicate technical ideas clearly, concisely and reproducibly
 - Reflect on limitations and scientific insight
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Submission Requirements

- **Length:** 6–10 pages (excluding references and appendix)
 - **Format:** PDF, single column, 11-pt font, 1-inch margins
 - **Code:** Public or private GitHub repository (link included)
 - **Reproducibility:** Deterministic run scripts and environment specification required. Your code should have accurate and easy-to-follow installation, training and evolution instructions to enable the TA to re-run your experiments.
 - **Submission:** Brightspace – Final Project Report
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Required Report Structure

1. Title & Team Information (Not graded)

- Clear, specific title (≤ 15 words)
- Group Number, Team member names and emails

2. Abstract (5 pts)

A concise summary of the project.

Should include:

- Problem and motivation
 - Research question(s)
 - Methods used
 - Key results and takeaways
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3. Introduction & Motivation (10 pts)

Builds on the **Motivation** section of your proposal.

Expectations:

- Clearly restate the problem and why it matters
 - Situate the work within DL/RL or the application domain
 - Define concrete stakes (metrics, users, systems)
 - Clearly stated and justified research question(s) or hypothesis.
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4. Related Work (10 pts)

Expanded version of the proposal's **Literature Gap** section.

Expectations:

- Summarize the most relevant prior work including citations (significant points will be deducted for miss, inaccurate and AI “hallucinated” citations)
 - Compare methods, assumptions, datasets, and evaluation protocols
 - Clearly articulate the gap your project addresses
 - Demonstrate understanding of where your work fits in the literature
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5. Methodology (25 pts)

Execution of the experimental plan proposed earlier. If space is a challenge, some information on hyperparameters, compute, and reproducibility can be moved to the appendix (include a footnote indicated that is where it will be found.)

Must include:

- **Environment / Dataset:** description and justification
 - **Models & Algorithms:** architectures, loss functions, update rules
 - **Baselines & Controls:** including random or heuristic baselines
 - **Hyperparameter Search:** search space and selection protocol
 - **Compute:** hardware, frameworks, versions
 - **Reproducibility:** seeds, run scripts, repository structure
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6. Experimental Results (20 pts)

Clear presentation of experimental findings.

Expectations:

- Results averaged over ≥ 10 seeds per condition (if less, please justify the choice.)
 - Appropriate plots (e.g., return vs episodes, steps or wall-clock time, accuracy, calibration, etc..)
 - Mean $\pm$ standard error reported
 - Comparisons under matched compute and evaluation protocols
 - Clear captions and interpretation of figures
 - A link or directions to access code for reproducible experiments
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7. Analysis & Discussion (15 pts)

Demonstrates scientific reasoning beyond raw results.

Expectations:

- Interpret why certain methods perform better or worse
 - Analyze failure cases and unexpected behaviors
 - Connect findings back to the research question
 - Discuss sensitivity, ablations, or qualitative insights
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8. Limitations, Risks, and Future Work (5 pts)

Critical reflection on the project.

Expectations:

- Honest discussion of limitations (compute, assumptions, choice of algorithm, stability, etc.)
 - What would be done differently with more time or resources
 - Plausible next research steps
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9. Conclusion (5 pts)

Clear and concise summary.

Expectations:

- Restate the research question
 - Summarize main findings
 - Emphasize key insight or contribution
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10. References (5 pts)

- Properly formatted citations. Select a citation format and use it consistently.
 - All referenced work must be cited
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Appendix (Optional)

- Additional plots, ablations, implementation details
 - Must be clearly labeled and referenced in the main text
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Grading Breakdown (100 pts)

Section	Points
Abstract	5
Introduction & Motivation	10
Related Work	10
Methodology	25
Experimental Results	20
Analysis & Discussion	15
Limitations & Future Work	5
Conclusion	5
References	5
Total	100

General Evaluation Criteria

Across all sections, the following will be assessed:

- **Scientific rigor:** sound experimental design and analysis
 - **Clarity:** well-organized, readable, and concise writing
 - **Reproducibility:** ability for others to re-run experiments
 - **Insight:** goes beyond “what happened” to explain “why”
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